

drones without needing a physical presence. InsideFPV's 'fly far' technology allows users to control drones over a range of 10 to 15 kilometres.

InsideFPV works across various sectors, including agriculture and defence. In agriculture, their drones are designed to spray pesticides and insecticides efficiently, significantly reducing the time and effort required by farmers. For instance, a task that would take a farmer two days can be completed in just two hours with the help of their drones. They claim that their technology can cover one acre of land in 12 minutes, and the drones are autonomous, allowing for easy operation by simply mapping the points via a mobile transmitter.

In the defence sector, InsideFPV produces drones for security, surveillance, and reconnaissance purposes. These drones are designed to be as stealthy as possible, making minimal noise while conducting missions, and can be operated remotely from anywhere in the world. The company has demonstrated this capability by flying drones from Bengaluru, the Netherlands, and Oman. "We are also developing thermal and surveillance drones specifically for defence purposes. One of our models, the Taste 250, can fly up to 250 kilometres per hour, making it nearly impossible to track using a mobile phone. This is part of our effort to push the boundaries of drone technology for enhanced security and operational efficiency," Arth explains.

InsideFPV acts as an original equipment manufacturer (OEM), designing and assembling drones in India. The startup has design patent rights, granted by the Indian Patent Office.

Arth explains that their claim to fame lies in the compact and lightweight design of their drones,



Arth Chowdhary, Founder, InsideFPV

making them easier to transport and deploy compared to competitor products. Their key selling points include their compactness, high speed, and innovative fly-far technology, which allows operators to control the drones remotely with a first-person view, enhancing the user experience and operational efficiency.

The company has also simplified the process of flying drones with its 'plug-and-fly' technology, where users only need to connect the battery and begin flying. This ease of use extends to their consumer-grade drones, such as the BIR V2 for STEM education and the Elevate V1, equipped with a 4K camera and designed for travel and journalism purposes.

Our consumer drones are priced to cater to a wide range of users, with prices ranging from ₹5000 to ₹300,000. Our drones start from ₹400,000 for agricultural purposes, while our defence drones, designed for advanced security applications, can cost up to ₹5 million, highlights Arth.

InsideFPV has also begun manufacturing lithium-ion batteries, although the cells themselves are still sourced from China due to the lack of manufacturing facilities in India.

The startup's primary source of revenue comes from selling drones, with sales occurring at a pace of one drone every three hours. In the last year alone, they sold over 1000 drones, a significant increase from the 800 sold in the previous year. They expect to generate revenue between sixty and seventy million this year and project exponential growth, with a target income of ₹2 billion by 2030. InsideFPV has also secured venture capital funding and plans to IPO in the next few years.

The company is expanding its manufacturing facility, currently 465 square metres (5000 square feet), with plans for an additional 1860 square metres (20,000 square feet) next year.

InsideFPV faces challenges in scaling its business, particularly in educating the public and training individuals to operate drones. Compared to countries like China, the lack of drone-specific education in India has been a significant hurdle. However, they are optimistic that by 2030, India will emerge as a global hub for drone technology, which aligns with the government's vision. "We also face challenges in sourcing components locally, particularly electronics, which has made us consider the option of in-house manufacturing if we can't find suitable partners. It is an obstacle we are working to overcome as we strive to maintain quality and innovation in our drones," concludes Arth.

As they look to the future, InsideFPV plans to launch new products, including a surveillance drone with 30x zoom, and expand their presence in the global market. They are also looking for component manufacturers, particularly in carbon fibre and electronics, who can help them scale their production. **EFY**

—Nitisha Dubey